Genuine Multipartite Entanglement Analysis

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Model
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Hubbard ring (1-D, PBC)

System parameters
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Sites (L)         : 6
Electrons (Ne)    : 6
Hopping (t)       : 1.0
Interaction (U)   : 4.0
Mode              : dimerized
n-body order      : 4
```

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Ground-state properties
-----------------------
E₀                : -2.48528137
E₁                : -1.65685425

⟨S²⟩              : 0.000000
S                 : 0.0000
⟨Sz⟩              : 0.000000
Multiplicity 2S+1 : 1.0
Spin state        : Singlet (S=0)

Double occupancy
----------------
⟨D⟩ total         : 0.439340
⟨D⟩ / site        : 0.073223
```

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GME summary
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Extracted n-spin RDM : n = 4

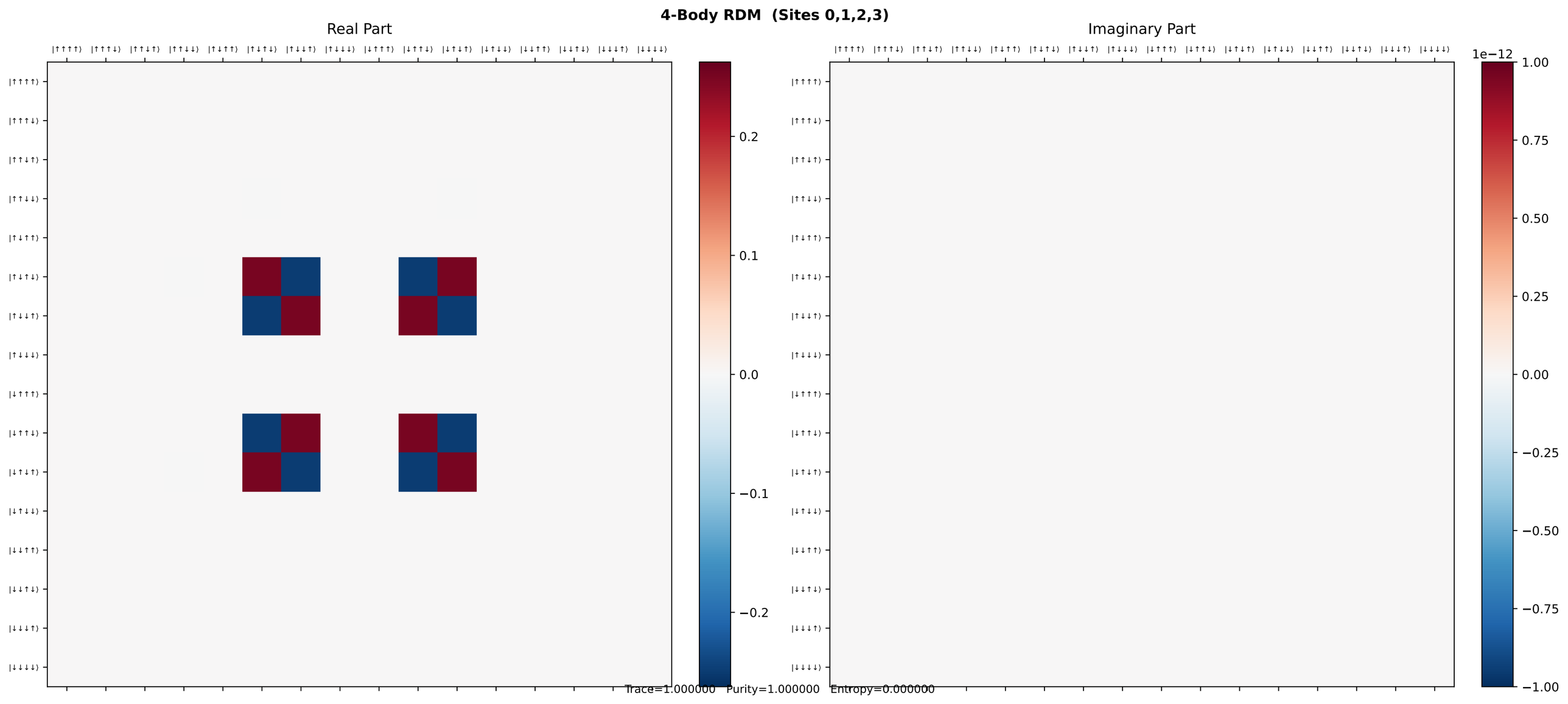
Purity:
  γ = Tr(ρ²)

Separable cut criterion:
  γ_A = 1  ↔  ρ = ρ_A ⊗ ρ_B

C_GME (Troiani et al.):
  C_GME = √(2 · min_A (1 − Tr(ρ_A²)))

Decision logic:
  γ = 1  and  all Tr(ρ_A²) < 1   → GME
  γ = 1  and  some Tr(ρ_A²) = 1  → No GME
  γ < 1                           → Mixed (convex-roof needed)

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4-Body RDM (Sites 0,1,2,3)
Real Part
Imaginary Part
1e−12
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000

```
RDM Metrics  (Sites = 0,1,2,3)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 1.0000000000
  Von Neumann entropy    : 0.0000000000
  Eigenvalue  min        : -7.445204e-17
  Eigenvalue  max        : 1.000000e+00
```

GME Analysis — 4-Body RDM (Sites 0,1,2,3)

Bipartitions (Sites 0,1,2,3)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 1.00000000$

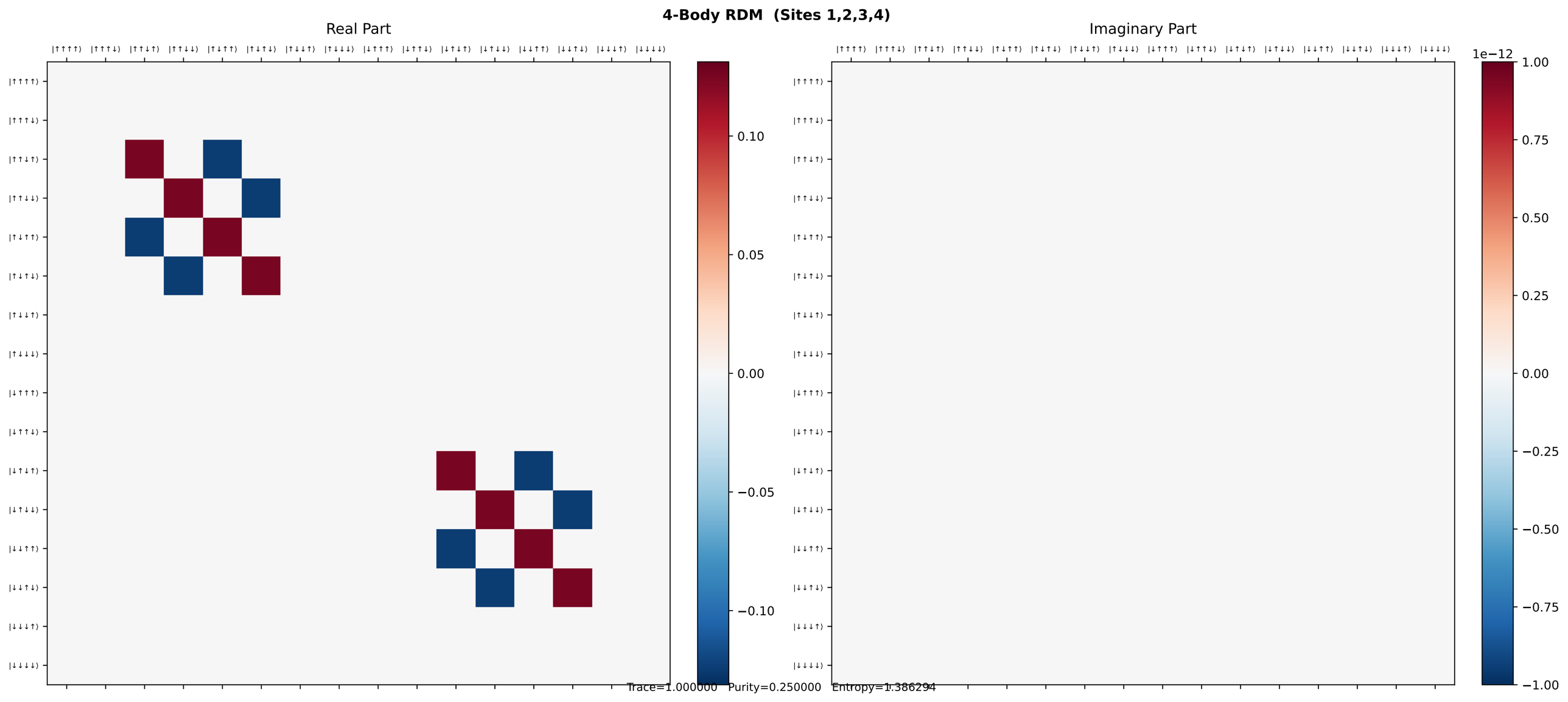
A	B	$\mathrm{Tr}(\rho_A^2)$	
(0,)	(1, 2, 3)	0.50000000	
(0, 1)	(2, 3)	1.00000000	←
(0, 2)	(1, 3)	0.25000000	
(0, 3)	(1, 2)	0.25000000	
(0, 1, 2)	(3,)	0.50000000	
(0, 1, 3)	(2,)	0.50000000	
(0, 2, 3)	(1,)	0.50000000	

STATUS: NO GME

At least one bipartition satisfies $\mathrm{Tr}(\rho_A^2) = 1 \rightarrow \rho = \rho_A \otimes \rho_B$

Separable cuts:

A=(0, 1) | B=(2, 3)


4-Body RDM (Sites 1,2,3,4)
Real Part
Imaginary Part
1e−12
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.250000 Entropy=1.386294

```
RDM Metrics  (Sites = 1,2,3,4)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 0.2500000000
  Von Neumann entropy    : 1.3862943611
  Eigenvalue  min        : -1.389857e-17
  Eigenvalue  max        : 2.500000e-01
```

GME Analysis — 4-Body RDM (Sites 1,2,3,4)

Bipartitions (Sites 1,2,3,4)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.25000000$

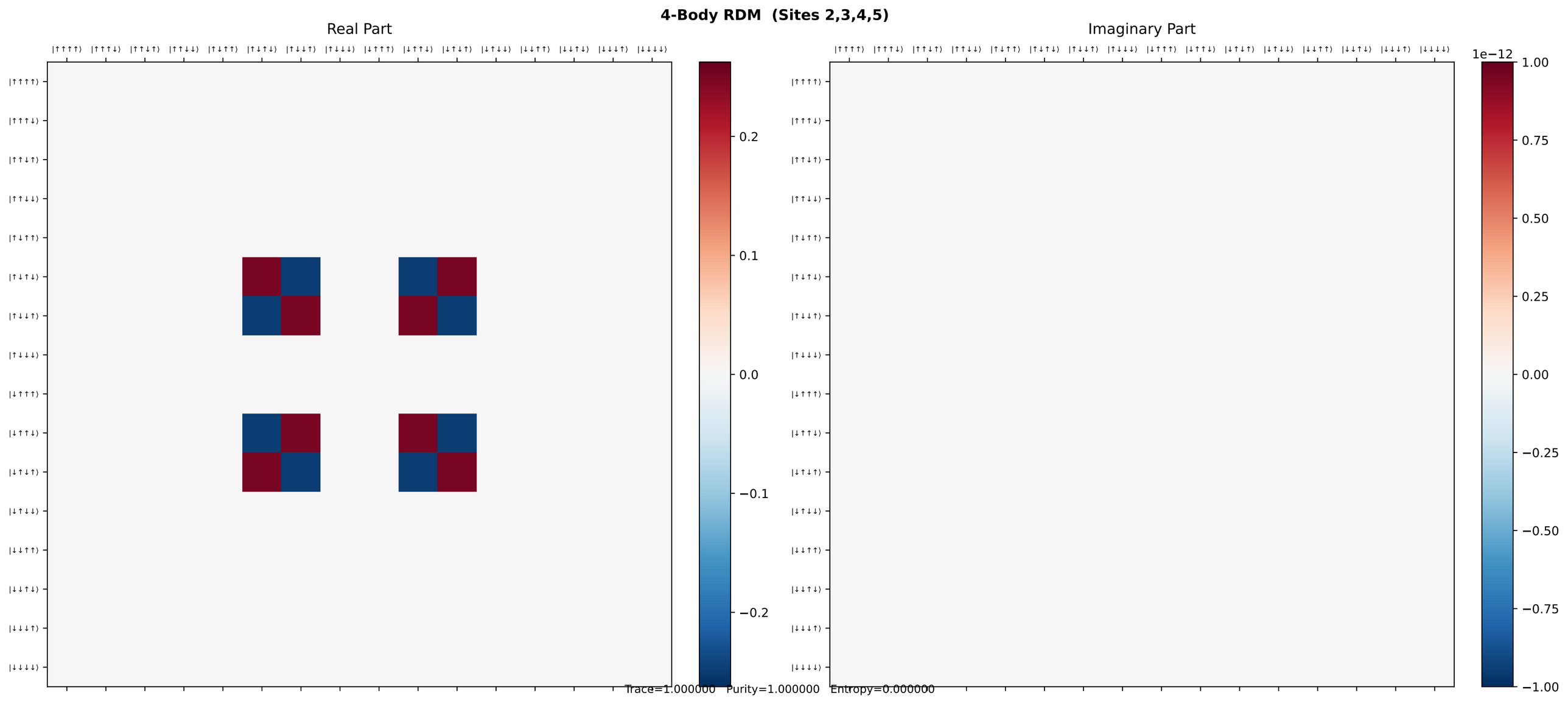
A	B	$\mathrm{Tr}(\rho_A^2)$

STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).


4-Body RDM (Sites 2,3,4,5)
Real Part
Imaginary Part
1e−12
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000

```
RDM Metrics  (Sites = 2,3,4,5)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 1.0000000000
  Von Neumann entropy    : 0.0000000000
  Eigenvalue  min        : -6.798700e-17
  Eigenvalue  max        : 1.000000e+00
```

GME Analysis — 4-Body RDM (Sites 2,3,4,5)

Bipartitions (Sites 2,3,4,5)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 1.00000000$

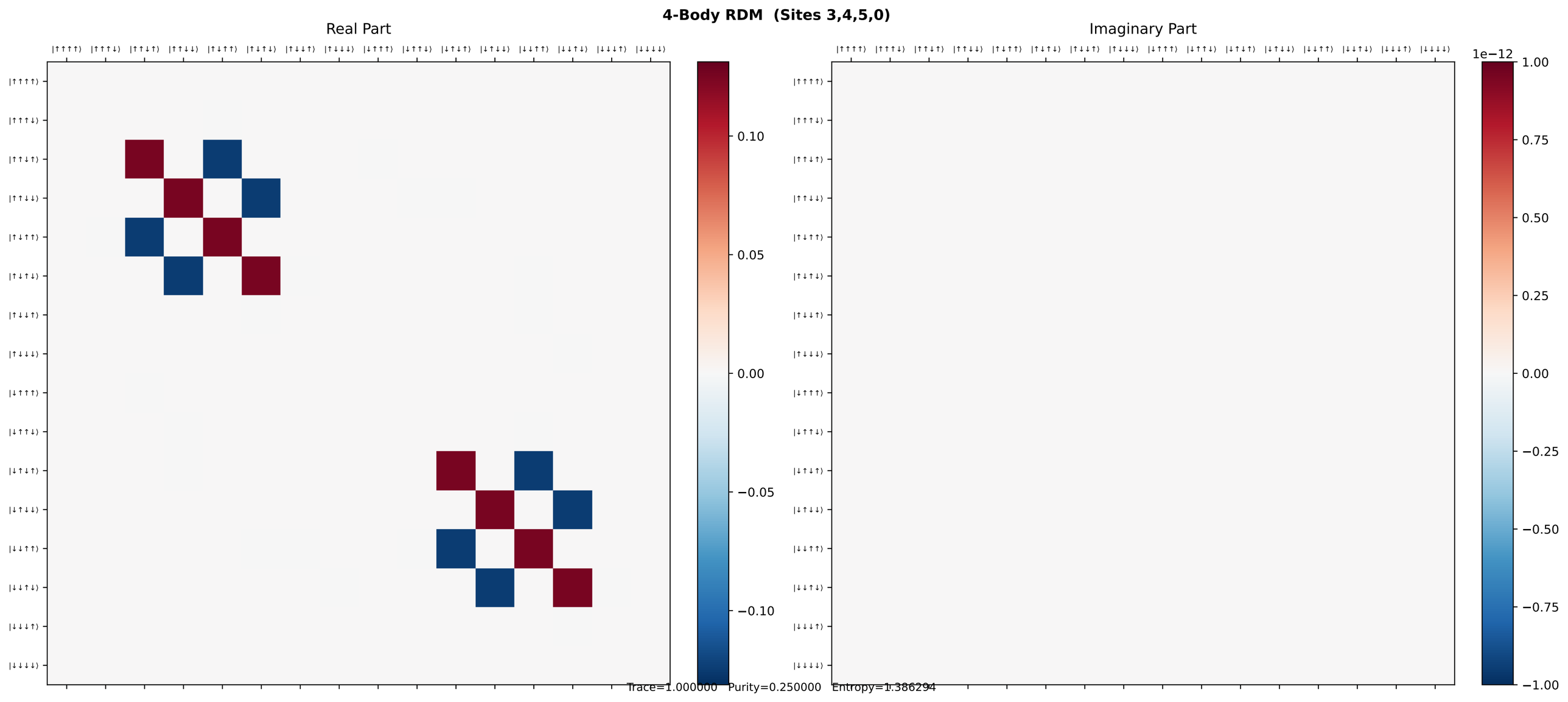
A	B	$\mathrm{Tr}(\rho_A^2)$	
(2,)	(3, 4, 5)	0.50000000	
(2, 3)	(4, 5)	1.00000000	←
(2, 4)	(3, 5)	0.25000000	
(2, 5)	(3, 4)	0.25000000	
(2, 3, 4)	(5,)	0.50000000	
(2, 3, 5)	(4,)	0.50000000	
(2, 4, 5)	(3,)	0.50000000	

STATUS: NO GME

At least one bipartition satisfies $\mathrm{Tr}(\rho_A^2) = 1 \rightarrow \rho = \rho_A \otimes \rho_B$

Separable cuts:

A=(2, 3) | B=(4, 5)


4-Body RDM (Sites 3,4,5,0)
Real Part
Imaginary Part
1e−12
Trace=1.000000 Purity=0.250000 Entropy=1.386294

```
RDM Metrics  (Sites = 3,4,5,0)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 0.2500000000
  Von Neumann entropy    : 1.3862943611
  Eigenvalue  min        : -6.937396e-17
  Eigenvalue  max        : 2.500000e-01
```

GME Analysis — 4-Body RDM (Sites 3,4,5,0)

Bipartitions (Sites 3,4,5,0)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.25000000$

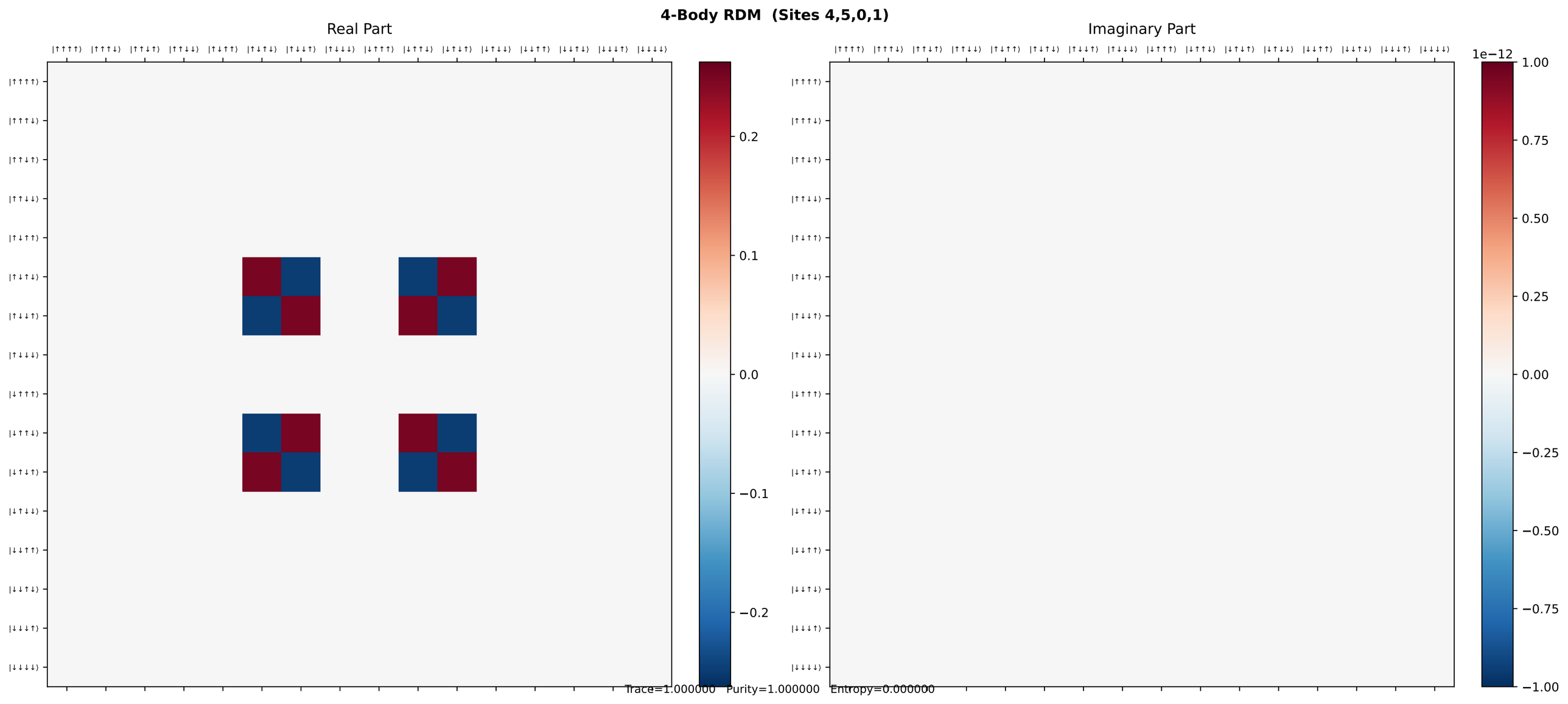
A	B	$\mathrm{Tr}(\rho_A^2)$

STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).


4-Body RDM (Sites 4,5,0,1)
Real Part
Imaginary Part
1e−12
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000

```
RDM Metrics  (Sites = 4,5,0,1)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 1.0000000000
  Von Neumann entropy    : 0.0000000000
  Eigenvalue  min        : -3.559881e-18
  Eigenvalue  max        : 1.000000e+00
```

GME Analysis — 4-Body RDM (Sites 4,5,0,1)

Bipartitions (Sites 4,5,0,1)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 1.00000000$

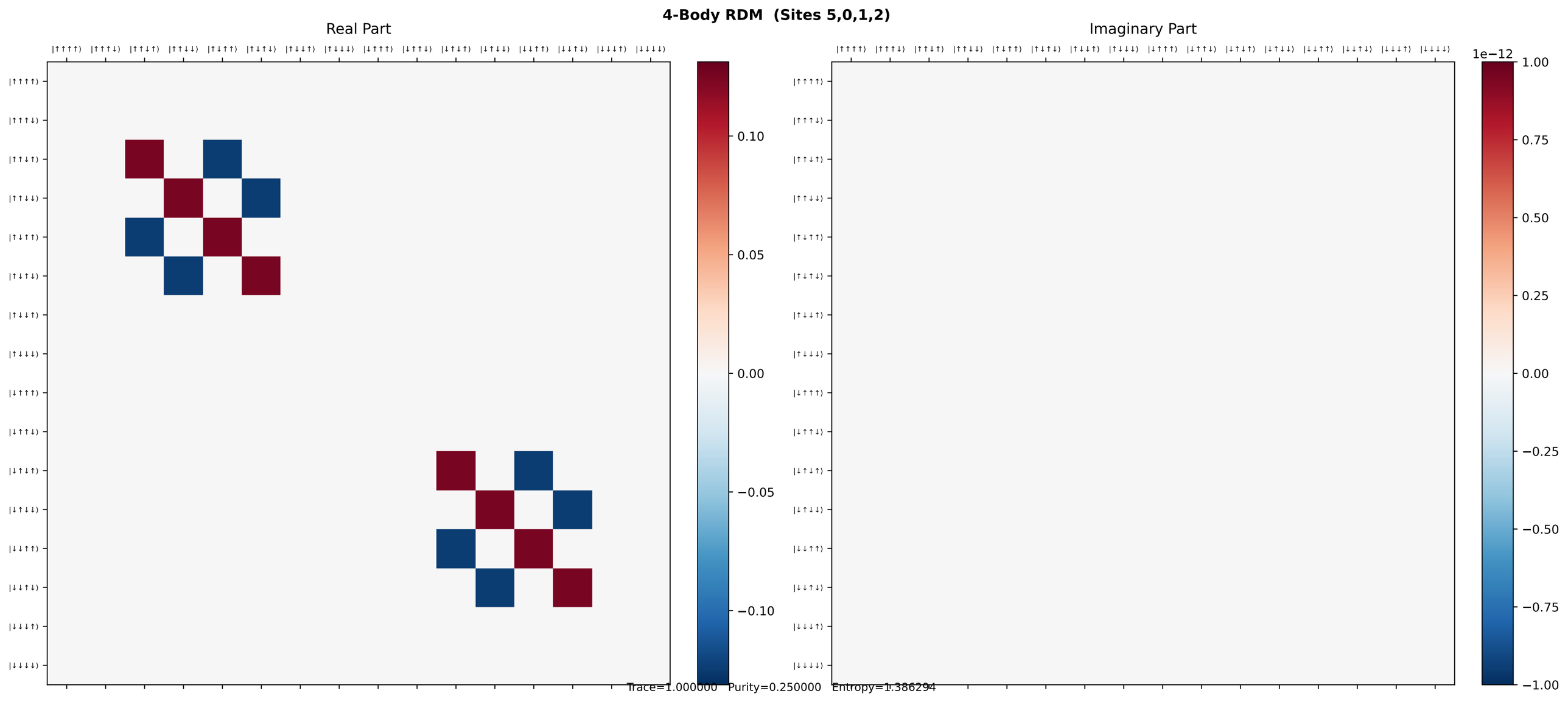
A	B	$\mathrm{Tr}(\rho_A^2)$	
(4,)	(5, 0, 1)	0.50000000	
(4, 5)	(0, 1)	1.00000000	←
(4, 0)	(5, 1)	0.25000000	
(4, 1)	(5, 0)	0.25000000	
(4, 5, 0)	(1,)	0.50000000	
(4, 5, 1)	(0,)	0.50000000	
(4, 0, 1)	(5,)	0.50000000	

STATUS: NO GME

At least one bipartition satisfies $\mathrm{Tr}(\rho_A^2) = 1 \rightarrow \rho = \rho_A \otimes \rho_B$

Separable cuts:

A=(4, 5) | B=(0, 1)


4-Body RDM (Sites 5,0,1,2)
Real Part
Imaginary Part
1e−12
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.250000 Entropy=1.386294

```
RDM Metrics  (Sites = 5,0,1,2)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 0.2500000000
  Von Neumann entropy    : 1.3862943611
  Eigenvalue  min        : -4.163488e-17
  Eigenvalue  max        : 2.500000e-01
```

GME Analysis — 4-Body RDM (Sites 5,0,1,2)

Bipartitions (Sites 5,0,1,2)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.25000000$

A	B	$\mathrm{Tr}(\rho_A^2)$

STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).